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Random Variables I: Introduction

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Stat 141 | Fall 2026

Friday, Week 10

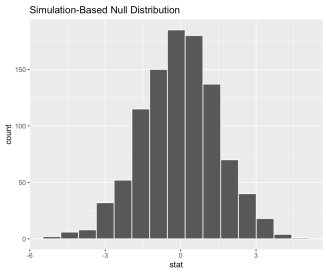
Goals for today

- Define and introduce **random variables**

Probability Models

Motivating question: How can we use theoretical probability models to approximate our (sampling) distributions?

Before we can answer that question and apply the models, we need to learn about the theoretical probability models themselves.



Random Variables

Definitions

A **random variable** is a numeric quantity whose value depends on the result of a random process.

- Use capital letters at the end of the alphabet (W, X, Y, Z) to denote **random variables**
- Use lowercase letters (w, x, y, z) to denote the **particular values** of a random variable
  - e.g., $X = x$ means "we observed the random variable X take the value x"
- Use equations to express **events** associated to random variables.
  - e.g., "$X = 5$" represents the event "The random variable X takes the value 5".
- Events associated to variables have probabilities of occurring.
  - e.g., $P(X = 5) = 0.5$ means X has probability 0.5 of taking the value 5.

Types of Random Variables

There are two main types of random variables:

1. **Discrete** variables can take only specific values (often a finite number of values).
2. **Continuous** variables can take any value in an interval of real numbers.

Examples of **discrete** variables:

- The number of credits a randomly chosen Reed student is taking.
- The number of vegetarians in a random sample of 10 people.
- The result of a coin flip

Examples of **continuous** variables:

- The temperature of my office at a particular time of the day.
- The amount of time it takes a radioactive particle to decay.

Today, we'll focus on discrete random variables

For a discrete random variable, care about its:

- Distribution: $p(x) = P(X = x)$
- Center (Mean):

$$\mu = \sum_x x p(x)$$

- Spread (Variance & Standard Deviation):

$$\sigma^2 = \sum_x (x - \mu)^2 p(x)$$

$$\sigma = \sqrt{\sum_x (x - \mu)^2 p(x)}$$

The Distribution of a Random Variable

Random variables have distributions, which tell us…

- the values the variable can take, and the *probability* the variable takes those values.

Example: I play a casino game, where that the amount of money I win (in dollars) has the following distribution:

Winnings	$5	$10	$20	$50
Probability	.3	.4	.2	.1

Suppose instead that I have a purse filled with the following 100 bills:

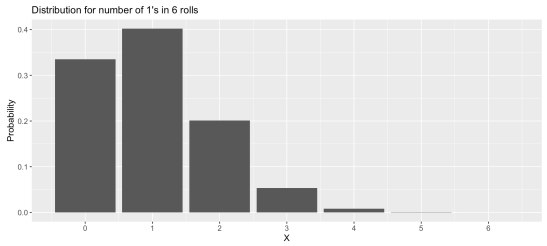
Type	$5	$10	$20	$50
Frequency	30	40	20	10

Playing this casino game is very similar to drawing a random bill from the purse!

Visualizing Discrete Distributions

We often use bar charts to visualize the distribution of discrete random variables.

- Suppose a fair 6-sided die is rolled 6 times. Let X be the number of 1s rolled. The distribution of X is given by:



- Heights of bars are **probabilities**
- This is analogous to rescaling a histogram to have heights = proportions instead of counts

Expected Value and Variance

Expected Value

Informally: The **expected value** is the average value the random variable takes.

- If X represents the number of Heads when flipping a fair coin, its expected value is 0.5

Formally: The **expected value** (or mean) of a discrete random variable X is

$$E[X] = x_1P(X = x_1) + x_2P(X = x_2) + \dots x_nP(X = x_n) = \sum_{i=1}^{n} x_iP(X = x_i)$$

where $x_1, \dots, x_n$ are *all the values X could potentially take*.

- The expected value of X is a **weighted average** of the values X can take, where weights are probabilities.
- We often use μ to denote expected values (i.e. μ = E[X])

Expected Value

Recall the example: if X represents the number of Heads when flipping a fair coin, its expected value is 0.5

$$\begin{aligned} E[X] &= 0P(X = 0) + 1P(X = 1) \\ &= 0(0.5) + 1(0.5) \\ &= 0.5 \end{aligned}$$

- The expected value of X is a **weighted average** of the values X can take, where weights are probabilities.

Practice

$$E[X] = x_1 P(X = x_1) + x_2 P(X = x_2) + \ldots x_n P(X = x_n) = \sum_{i=1}^{n} x_i P(X = x_i)$$

Suppose we have a data set consisting of 10 values: $\{1, 1, 2, 2, 2, 2, 3, 4, 5, 5\}$.

- Let X be a value chosen from this data set randomly.
- What is the expected value of X?

02:00

The Law of Large Numbers, again

Previously, we said that by the **Law of Large numbers** (LLN), the proportion of times an outcome occurs in a long sequence of trials is close to the probability for that outcome.

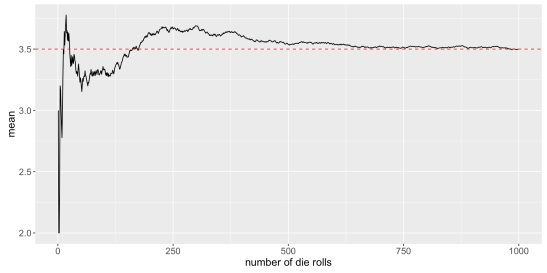
This is a generalization:

> **Theorem: The Law of Large Numbers**
>
> Let X be a random variable.
>
> 1. Suppose we observe the random variable n times, $x_1, x_2, \ldots, x_n$.
> 2. Let $\bar{x}_n$ denote the mean of our n observations of X.
> 3. Then, as n becomes large, $\bar{x}_n$ will approach the expected value $E[X]$ of the random variable X.

A Roll of the Die

Suppose we roll a fair 6-sided die. What is the expected value of the result?

- Let's roll the same die 1000 times and keep track of the running mean of the results…



We can see that the **expected value** is 3.5

Variance and Standard Deviation

The **variance** of a discrete random variable X with mean $E(X) = \mu$ is

$$\begin{aligned}\text{Var}(X) =& (x_1 - \mu)^2 P(X = x_1) + (x_2 - \mu)^2 P(X = x_2) + \cdots + (x_n - \mu)^2 P(X = x_n) \\ =& \sum_{i=1}^{n} (x_i - \mu)^2 P(X = x_i)\end{aligned}$$

The variance of X is the **sum the squared deviations of X from its mean μ, weighted by the corresponding probabilities.**

We also define the **standard deviation** of a random variable X to be

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

We often use:

- σ^2 to denote the variance of a variable
- σ to denote the standard deviation

Variance and Standard Deviation

$$\begin{aligned}\text{Var}(X) &= (x_1-\mu)^2P(X=x_1)+(x_2-\mu)^2P(X=x_2)+\cdots+(x_n-\mu)^2P(X=x_n)\\ &= \sum_{i=1}^{n}(x_i-\mu)^2P(X=x_i)\end{aligned}$$

Suppose we have a data set consisting of 5 values: $\{1,1,3,5,5\}$. Let X be a value chosen from this data set randomly. What is the variance of X?

- Note that $E(X)=\mu=3$

03:00

Practice: One Coin Flip

Consider a random variable X which is the number of heads in a single **fair** coin flip.

Q: What are the possible values for X? What are the probabilities each value of X?

- $X=1$ (H) with $P(X=1)=\frac{1}{2}$
- $X=0$ (T) with $P(X=0)=\frac{1}{2}$

Q: Compute the expected value and variance for X (the number of heads in a single coin flip)

$$E[X]=(0)\frac{1}{2}+(1)\frac{1}{2}=\frac{1}{2}$$

$$Var[X]=(0-1/2)^2\frac{1}{2}+(1-1/2)^2\frac{1}{2}=\frac{1}{4}$$

Linearity of Expected Value

Theorem: Expectation of Sum

Let X and Y be random variables. Then

$$E(X+Y)=E(X)+E(Y)$$

- e.g., If X is the face of one die roll, and Y is the face of another die roll, then $X+Y$ would be the sum of the faces. And:

$$E(X+Y)=E(X)+E(Y)=3.5+3.5=7$$

Linearity of Expected Value

Theorem: Scalar Multiplication with Expectation

Let X be a random variable, and let c be a number. Then

$$E(cX)=cE(X)$$

- e.g., We're rolling a die, and we'll make $5 times the number we roll – what are our expected earnings?

$X=$ face of die roll $\qquad E(5X)=5E(X)=5(3.5)=17.5$

Variance of Sums and Scalar Multiplication

Theorem: Variance of Sum

Let X and Y be random variables. Additionally, let X and Y be *independent*. Then

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

- Note that X and Y **must be independent** for the above!

Theorem: Scalar Multiplication with Variance

Let X be a random variable, and let c be a number. Then

$$\text{Var}(cX) = c^2\text{Var}(X)$$

- We get the c^2 because variances consider squared differences from the mean

Practice: Two Coin Flips

Now let Y be the number of heads in two coin flips.

Q: What are the possible values for Y? What are the probabilities each value of Y?

- $Y = 0$ (TT) with $P(Y=0) = \frac{1}{4}$
- $Y = 1$ (HT or TH) with $P(Y=1) = \frac{1}{2}$
- $Y = 2$ (HH) with $P(Y=2) = \frac{1}{4}$

Q: Compute the expected value and variance for Y (the number of heads in two coin flips.)

We could use the definitions of expected value and variance directly…

$$E[Y] = (0)\frac{1}{4} + (1)\frac{1}{2} + (2)\frac{1}{4} = 1$$

$$Var[Y] = (0-1)^2\frac{1}{4} + (1-1)^2\frac{1}{2} + (2-1)^2\frac{1}{4} = \frac{1}{2}$$

Practice: Two Coin Flips

…or, we could use our rules for expected value and variance:

$$Y = X_1 + X_2$$

where X_1 and X_2 are two independent versions of our random variable, X.

Thus,

$$E[Y] = E[X_1] + E[X_2] = 0.5 + 0.5 = 1$$

$$Var[Y] = Var[X_1] + Var[X_2] = 0.25 + 0.25 = 0.5$$

That's a lot simpler!

Recap: Properties of Random Variables

The **expected value** (or mean) of a discrete random variable X is

$$E[X] = \mu = x_1P(X=x_1) + x_2P(X=x_2) + \ldots x_nP(X=x_n) = \sum_{i=1}^{n} x_iP(X=x_i)$$

where $x_1, \ldots, x_n$ are *all the values X could potentially take*.

The **variance** of a discrete random variable X with mean $E(X) = \mu$ is

$$\text{Var}(X) = \sigma^2 = \sum_{i=1}^{n}(x_i-\mu)^2P(X=x_i)$$

We also define the **standard deviation** of a random variable X to be

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Recap: Theorems for Discrete Random Variables

Theorem: Expectation of Sum

Let X and Y be random variables. Then

$$E(X+Y) = E(X) + E(Y)$$

Theorem: Scalar Multiplication with Expectation

Let X be a random variable, and let c be a number. Then

$$E(cX) = cE(X)$$

Theorem: Variance of Sum

Let X and Y be random variables. Additionally, let X and Y be ***independent***. Then

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

Theorem: Scalar Multiplication with Variance

Let X be a random variable, and let c be a number. Then

$$\text{Var}(cX) = c^2\text{Var}(X)$$

Next time:

- We'll define some specific named random variables!